EXP.NO:07

FLOW CONTROL AT DATA LINK LAYER USING SLIDING WINDOW PROTOCOL

AIM:

To write the program to implement flow control at the Data link layer using Sliding Window Protocol, and to simulate the flow of frames from one node to another.

ALGORITHM:

* Start the program.
* Read total frames N, window size w & lost frame.
* Initialize base -0 and nextSeq - 0
* Repeat untill all frames are transmitted.
  - Create the current sender windows.
  - Send all frames in the window.
  - If a frame is lost, (Go - Back - N).
  - Otherwise, receive ACK's & slide the window forward.
* Repeat untill all frames are successful.
* Stop the program.

CODE:

```
def simulate (n, w, lost):
    base = 0
    while base < n:
        end : min (base + w, n)
        print (f "\n Window: [{base} - {end - 1}]")
        for i in range (base, end):
            print (f "A → Sending Frame {i}")
        if i == lost :
            print (f"Frame {i} LOST")
            print (f "B → Frame {i} missing")
            print (f"A → Timeout. resending from {i}")
            lost = -1
            break
```

```
        else:
            for i in range(base, end):
                print(f"B → Received Frame {i}, ACK {i}")
            base = end
print("\n All frames transmitted successfully")
n = int(input("Total Frames: "))
w = int(input("Window Size: "))
lost = int(input("Lost Frame (-1 for none): "))
simulate(n, w, lost)
```

RESULT:

Thus, the Sliding Window Protocol using Go-Back-N was successfully implemented.

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send      : 10
Enter sender's window size                : 3
Enter frame number to simulate as LOST (0 to 9, enter -1 for no loss): 2

-----------------------------------------------
Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
        (Frame 2 LOST in transit!)
Node B --> Frame 2 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 2.

-----------------------------------------------
Sender Window : [Frame 0 - Frame 2]
Node A --> Sending Frame 2
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node A --> ACK 2 received. Sliding window forward.

-----------------------------------------------
Sender Window : [Frame 3 - Frame 5]
Node A --> Sending Frame 3
Node A --> Sending Frame 4
Node A --> Sending Frame 5
Node B --> Received Frame 3, sending ACK 3
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5

Node A --> ACK 5 received. Sliding window forward.

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Sender Window : [Frame 6 - Frame 8]
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node A --> Sending Frame 8
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node B --> Received Frame 8, sending ACK 8
Node A --> ACK 8 received. Sliding window forward.

-----------------------------------------------
Sender Window : [Frame 9 - Frame 9]
Node A --> Sending Frame 9
Node B --> Received Frame 9, sending ACK 9
Node A --> ACK 9 received. Sliding window forward.

-----------------------------------------------
All 10 frames successfully transmitted from Node A to Node B.
```